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Medicare Part D Plans Greatly Increased Utilization Restrictions On Prescription Drugs, 2011–20

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ABSTRACT Drug utilization management tools can be employed to ensure that medicines are prescribed cost-effectively, but they can also be implemented in ways that reduce adherence and harm patient health. We examined trends in the prevalence of utilization restrictions on non-protected-class compounds in Medicare Part D plans during the period 2011–20, including prior authorization and step therapy requirements as well as formulary exclusions. Part D plans became significantly more restrictive over time, rising from an average of 31.9 percent of compounds restricted in 2011 to 44.4 percent restricted in 2020. The prevalence of formulary exclusions grew particularly fast: By 2020, plan formularies excluded an average of 44.7 percent of brand-name-only compounds. Formulary restrictions were more common among brand-name-only compared with generic-available compounds, among more expensive compounds, and in stand-alone compared with Medicare Advantage prescription drug plans.

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Pharmacy benefit managers frequently use tools such as prior authorization and step therapy requirements to manage the use of medicines among their beneficiary populations. They typically justify the use of these tools as mechanisms to control costs and enhance safety by ensuring that patients do not use more expensive treatments when less expensive but equally effective therapies are available. Over time, pharmacy benefit managers have employed these controls more liberally, placing them on a steadily increasing share of compounds, and research has documented that prior authorization and step therapy are associated with decreased use of the restricted drugs.[1]

There has also been a steady increase during the past decade in the number of compounds excluded from plan formularies.[2] Unlike prior authorization or step therapy restrictions, which require additional administrative steps that may delay a patient's eventual access to a restricted drug, exclusions or so-called closed formularies remove the drug from coverage altogether, foreclosing access unless a patient can pay the full cost out of pocket or successfully appeal.[3] Thus, exclusions are a more extreme mechanism for reducing plan drug costs, which potentially harm patients by restricting needed care.

We examined trends in the prevalence of prior authorization or step therapy and formulary exclusions in Medicare during 2011–20, finding increasing restrictiveness in Part D plans over time. We also documented differences in restrictiveness across brand-name-only versus generic-available drug categories, across average prescription costs, and between Medicare Advantage prescription drug (MA-PD) plans and stand-alone prescription drug plans.

Study Data And Methods

DATA This retrospective study used the 100 percent sample of Medicare administrative data for

2011–20, including plan and formulary characteristics, beneficiary enrollment, and pharmacy (Part D) claims. Medicare Part D offers two ways to obtain prescription drug coverage: stand-alone prescription drug plans, which add separate drug coverage to traditional fee-for-service Medicare, and MA-PD plans, which integrate drug coverage into a Medicare Advantage health maintenance organization or preferred provider organization plan. Because MA-PD plans are at financial risk for both medical and drug expenditures, they face different incentives than stand-alone prescription drug plans in restricting access to medications that could affect the use of medical services. Formulary characteristics included Chronic Conditions Data Warehouse (CCW) sequence numbers analogous to National Library of Medicine RxNorm Concept Unique Identifiers, FirstDataBank brand name and proprietary clinical formulation identifiers, utilization management restrictions such as step therapy and prior authorization, and an indicator of whether the product was available over the counter. Part D claims included the CCW sequence numbers, National Drug Codes, fill dates, days supplied, and total cost of the prescriptions.

DRUG COMPOUNDS AND FORMULARY RESTRICTIONS The analysis was conducted at the drug compound (that is, generic name) level.[4] FirstDataBank data, in combination with the formulary characteristics and Part D claims data, were used to harmonize generic names across years; some manual matching was required. The FirstDataBank data also contained information on therapeutic class, an indicator of generic or brand-name product, and the product marketing date.

Medicare Part D plans are required to cover all or substantially all drugs in six protected classes—anticonvulsants, antidepressants, antineoplastics, antipsychotics, antiretrovirals, and immunosuppressants—and are limited in how they may use prior authorization or step therapy for these drugs.[5,6] Therefore, the universe of compounds was defined as those covered by any formulary, excluding over-the-counter products and protected classes.

Compounds were classified as “brand-name only” (that is, single-source brand) if no generic version existed in that year, based on the FirstDataBank definitions. All other compounds were classified as “generic available” in a given year; products within such compounds could include both generics and multisource brands. Compounds were further categorized by plan-year, subject to certain constraints on the compound’s use. A compound was categorized as “excluded” if no product within that compound was covered by the plan. Further, a compound was categorized as “under prior authorization or step therapy” only if all covered products within that compound required prior authorization or step therapy. For example, the cholesterol-lowering compound atorvastatin would not be under prior authorization or step therapy if either the generic or the brand-name version (Lipitor) was available without restrictions.[7]

ANALYSIS The fraction of compounds categorized as excluded or under prior authorization or step therapy was weighted by plan enrollment and calculated separately for brand-name-only and generic-available compounds. Weighting the prevalence of prior authorization or step therapy and formulary exclusions by plan enrollment measures the share of Part D beneficiaries exposed to these restrictions but treats each compound equally, regardless of how widely it is used. To assess the share of drug spending affected by these restrictions, we also weighted by average per member spending in plans that did not impose any restrictions on that compound.[8] Results using these alternative weights are in the online appendix.[9] To determine the association between the cost of a drug and formulary restrictions, we categorized compounds into five cost categories (up to \$100, \$101–\$250, \$251–\$500, \$501–\$1,000, and more than \$1,000), based on the average total cost per prescription. The analysis is also presented separately for stand-alone versus MA-PD plans. All analyses were conducted using SAS, version 9.4.

LIMITATIONS Our study faced several limitations. We did not consider how prior authorization or step therapy requirements are enforced: Patients may still be granted access to compounds under prior authorization or step therapy, and denials may be overturned on appeal. Our analysis could not assess the extent to which patients were harmed by these policies. We also did not distinguish between utilization restrictions that may be justified on a cost-effectiveness basis and those that save costs but potentially harm patient health.

Study Results

Between 2011 and 2020, the number of plans, formularies, and beneficiaries in Part D all increased, rising from 33.2 million beneficiaries in 4,179 plans using 301 formularies in 2011 to 52.0 million beneficiaries in 6,160 plans using 544 formularies in 2020 (appendix exhibit A1).[9] MA-PD plans made up an increasing share of plans, from 68.7 percent of all Medicare plans in 2011 to 80.2 percent in 2020. MA-PD plans were smaller than stand-alone prescription drug plans, with the average MA-PD plan covering

5,001 beneficiaries in 2020, compared with 22,105 beneficiaries in the average stand-alone plan in that same year.

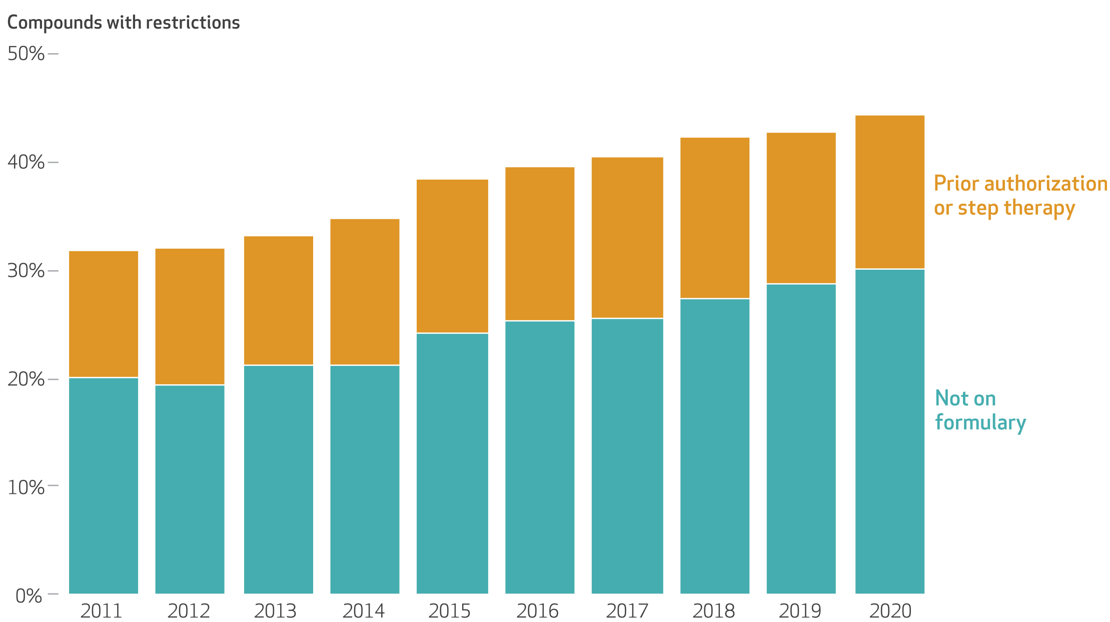
Formulary exclusions and prior authorization or step therapy restrictions affected an increasing share of compounds over the course of the period studied (exhibit 1). In 2011, plans excluded, on average, 20.4 percent of compounds from their formularies and subjected another 11.5 percent to prior authorization or step therapy restrictions. By 2020, plans excluded 30.4 percent of compounds and placed another 14.0 percent under prior authorization or step therapy. Restrictions increased over time for both brand-name-only and generic-available compounds, but the frequency of exclusions and prior authorization or step therapy restrictions was greater in all years for brand-name-only compounds (exhibit 2). In 2020, plans excluded, on average, 21.9 percent of compounds with a generic available and placed another 8.1 percent under prior authorization or step therapy. Among brand-name-only compounds, plans excluded 44.7 percent and subjected an additional 23.7 percent to prior authorization or step therapy restrictions, meaning that 68.4 percent of brand-name-only compounds faced some sort of utilization restriction in 2020. (See appendix exhibit A8 for data underlying exhibits 1 and 2.)[9]

The degree of access restriction varied by both the generic availability and the cost of the compound (exhibit 3). In 2020, 16.7 percent of generic-available compounds with an average cost of $100 or less per script were either excluded or under prior authorization or step therapy, compared with 59.5 percent of generic-available compounds with an average cost above $1,000. Among brand-name-only compounds, 15.8 percent of those costing $100 or less faced restrictions, whereas 83.7 percent of those costing more than $1,000 did. In almost all cost categories, exclusions restricted a larger share of compounds than prior authorization or step therapy, but the relationship was not monotonic in price: The largest shares of excluded compounds of both types were for drugs costing $251–$500. More expensive compounds had a smaller share excluded. (See appendix exhibit A8 for the data underlying exhibit 3.)[9]

Utilization restrictions were greater among stand-alone prescription drug plans than MA-PD plans across all years, and they increased over time (exhibit 4). In 2020, among brand-name-only compounds, stand-alone plans excluded 48.1 percent of compounds, on average, and restricted another 23.6 percent through prior authorization or step therapy requirements, whereas MA-PD plans excluded 41.0 percent of

EXHIBIT 1

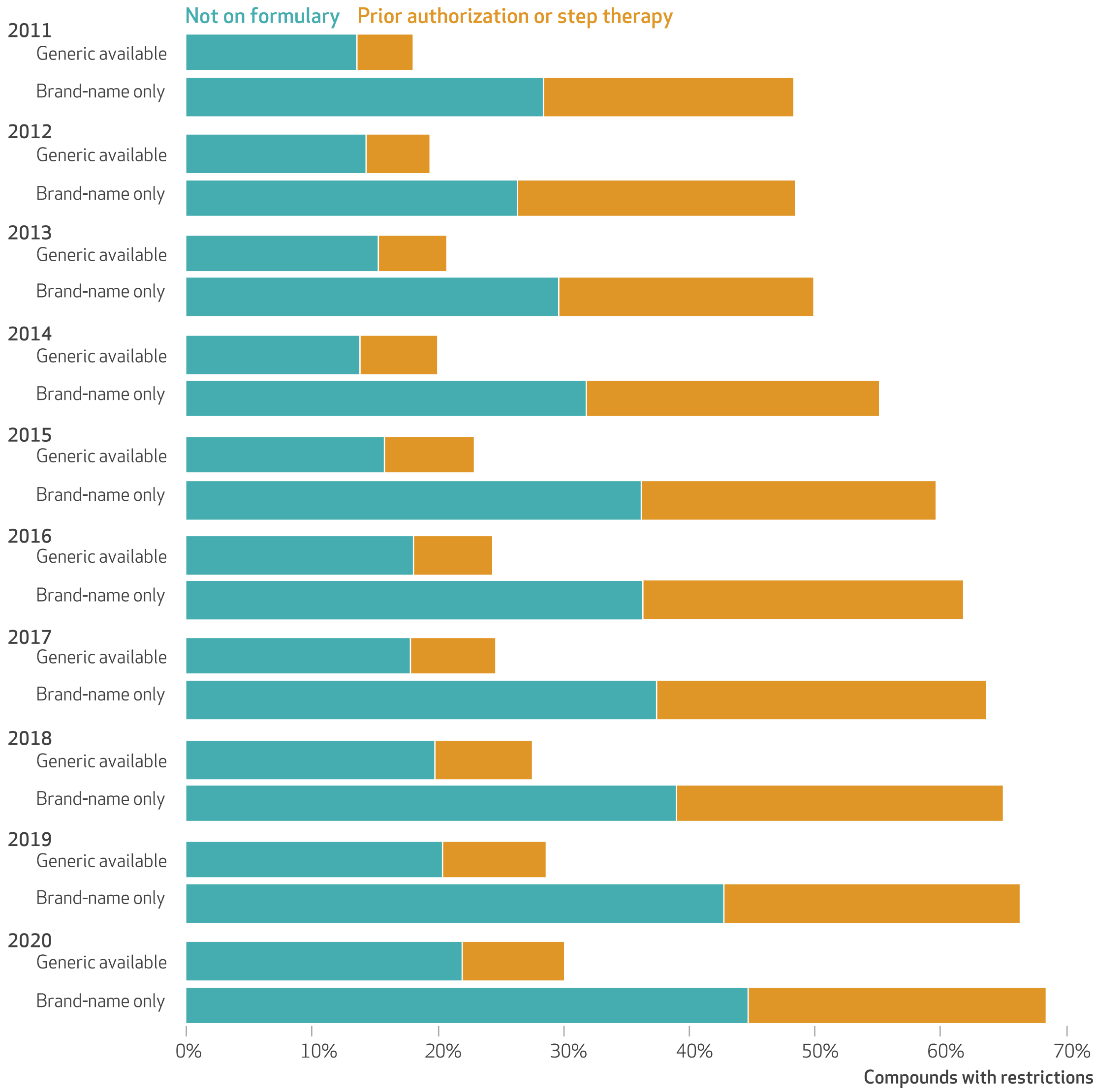
Proportion of compounds with restrictions (not on formulary, or prior authorization or step therapy) on Medicare Part D formularies, 2011–20



SOURCE Authors' analysis of data from the 100 percent sample of Medicare administrative data, including Part D formularies files, 2011–20. **NOTES** The universe of compounds was defined as those covered by any formulary, including excluded drugs if a plan covered excluded drugs and excluding over-the-counter products and protected classes. Averages are weighted by plan enrollment.

EXHIBIT 2

Proportion of compounds with restrictions (not on formulary, or prior authorization or step therapy) on Medicare Part D formularies, by generic availability, 2011–20



SOURCE Authors' analysis of data from the 100 percent sample of Medicare administrative data, including Part D formularies files, 2011–20. **NOTES** The universe of compounds was defined as those covered by any formulary, including excluded drugs if a plan covered excluded drugs and excluding over-the-counter products and protected classes. Averages are weighted by plan enrollment.

compounds and applied prior authorization or step therapy restrictions to another 23.9 percent. Although restrictions were generally less prevalent among generic-available compounds, they demonstrated similar patterns (appendix exhibit A2).[9]

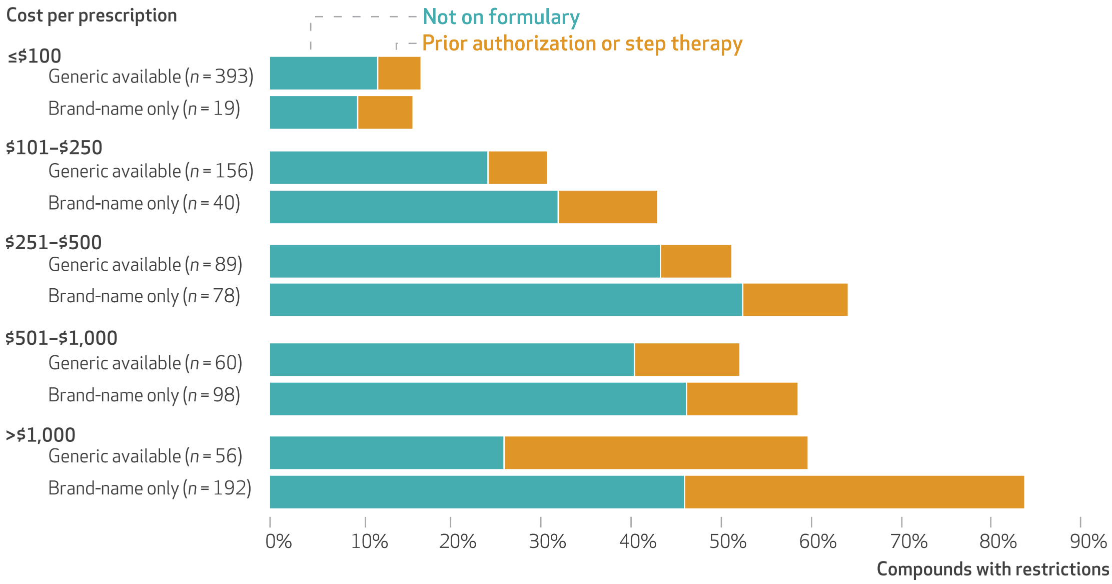
We replicated these analyses using an alternative weighting scheme that weighted each compound by per member sales, rather than plan enrollment (see appendix exhibits A3–A7).[9] Compared with beneficiary-weighted results, increasing trends in exclusions and prior authorization or step therapy were attenuated when we used weighting by sales, but the patterns were similar. Most of the increase was driven by greater use of formulary exclusions, particularly in stand-alone prescription drug plans.

Discussion

Prescription drug insurance has become an increasingly important element of the Medicare system, helping millions of older Americans afford their medications. Between 2011 and 2020, the total number of beneficiaries with Medicare Part D insurance grew by 56 percent, going from

EXHIBIT 3

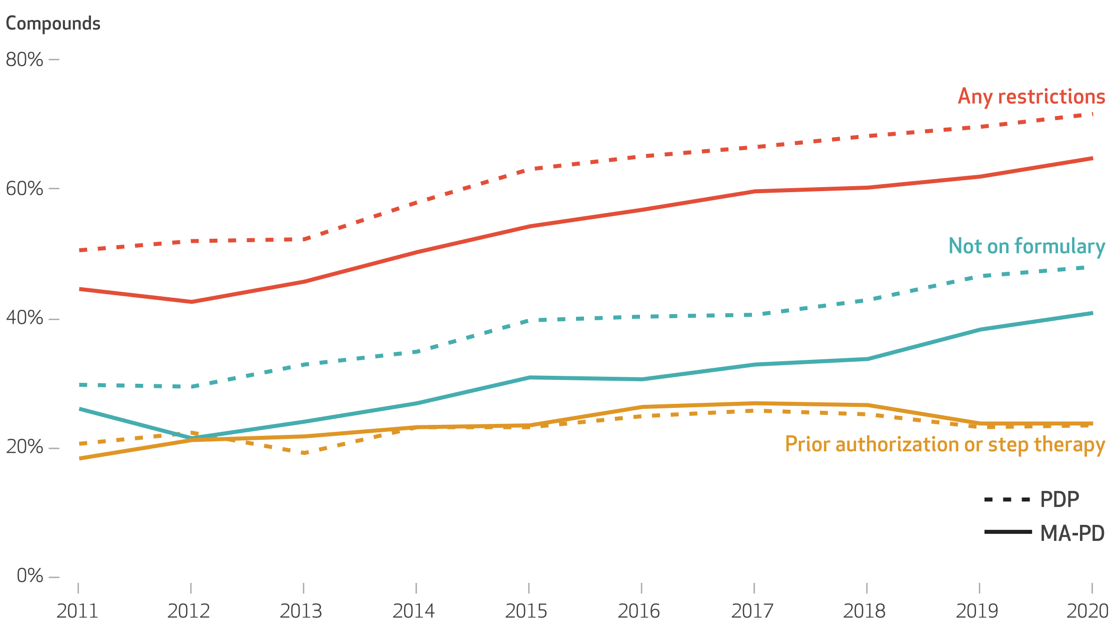
Proportion of compounds with restrictions (not on formulary, or prior authorization or step therapy) on Medicare Part D formularies, by mean cost per prescription in 2020



SOURCE Authors' analysis of data from the 100 percent sample of Medicare administrative data, including Part D formularies files, 2011–20. **NOTES** Cost ranges reflect the mean cost of a prescription in claims in 2020. Compounds without cost information in the claims are not shown in the exhibit (generic available: $n = 6$, brand-name only: $n = 28$). The universe of compounds was defined as those covered by any formulary, including excluded drugs if a plan covered excluded drugs and excluding over-the-counter products and protected classes. Mean costs are weighted by plan enrollment.

EXHIBIT 4

Comparison of restrictions (not on formulary, or prior authorization or step therapy) in Medicare Part D stand-alone prescription drug plans (PDPs) and Medicare Advantage Prescription Drug (MA-PD) plans on compounds available as brand-name only, 2011–20



SOURCE Authors' analysis of data from the 100 percent sample of Medicare administrative data, including Part D formularies files, 2011–20. **NOTES** The universe of compounds was defined as those covered by any formulary, including excluded drugs if a plan covered excluded drugs and excluding over-the-counter products and protected classes. Averages are weighted by plan enrollment.

33.2 million in 2011 to 52.0 million in 2020 (see appendix exhibit A1).[9] But as the number of people covered has increased, the coverage itself has narrowed. Part D formularies have become less inclusive, such that in 2020, three in ten generic-available compounds and more than two-thirds of brand-name-only compounds required prior authorization or step therapy or were excluded from coverage altogether, when weighted by enrollment (exhibit 2).

Although it is common, the use of formulary exclusions and prior authorization or step therapy restrictions is controversial. Health plans and pharmacy benefit managers contend that these strategies reduce spending and unnecessary utilization by steering patients to drugs that carry a lower net cost to the plan. However, physicians often object that these policies are increasingly overused, impose an administrative burden, and undermine their clinical decision making.[10]

In addition, physicians, pharmaceutical manufacturers, and patient advocates argue that these policies can induce patients to delay treatment, substitute less effective medications, or fail to comply with prescribed regimens, resulting in adverse health effects. Studies of utilization restrictions have shown both benefits and harms,[11–16] often varying by the essentiality of the drug class, the availability of safe and effective substitute drugs, and variability in administrative barriers to restricted agents.

Although encouraging the use of less expensive alternatives is often warranted, the large and increasing number of compounds that are subject to review or excluded from coverage raises concerns of the impact on patient health. Historically, exclusions were typically applied to drugs with generic equivalents or classes with multiple treatments of similar effects. However, at this time, exclusions are increasingly being applied to drugs for complex conditions such as cancers and autoimmune disorders that are often characterized by heterogeneous treatment effects across products and patients.[17,18] Further, because different pharmacy benefit managers exclude different medicines and often change their formularies from year to year, Medicare beneficiaries must navigate an ever-changing benefit landscape. This is particularly challenging if a beneficiary switches plans or their plan changes pharmacy benefit managers. Such complexity can exacerbate disparities in medication use and disadvantage sicker patients who take more medications.[19]

A recent analysis of prior authorization requirements compared the savings from changing medication use to the administrative cost borne by providers.[1] The study focused on Medicare beneficiaries receiving low-income subsidies who did not actively choose a plan and thus were randomly assigned to a Part D plan. The researchers found that prior authorization reduced the use of the targeted drug by 26.8 percent, generating savings for Medicare that exceeded the costs of administering prior authorization. Concerningly, however, only half of patients facing prior authorization switched to a substitute product; the remainder took no substitute drug. The very large increases that we found in formulary exclusions and prior authorization or step therapy during 2011–20 justify further research on the intended and unintended effects of such policies.

Our finding that MA-PD plans generally impose fewer formulary restrictions than stand-alone prescription drug plans is somewhat contrary to perceptions that managed care organizations impose more constraints on utilization. However, unlike stand-alone plans, MA-PD plans are also responsible for their beneficiaries' medical claims, so if an expensive drug can keep a patient out of the hospital or reduce the need for outpatient care or expensive procedures, MA-PD plans have a stronger financial incentive to provide access to it. Another possibility is that managed care organizations routinely employ other utilization management techniques to monitor and influence the prescribing behavior of their providers. They thus may rely more on metrics and peer comparisons than explicit formulary design to control utilization. The narrower coverage by stand-alone prescription drug plans compared with MA-PD plans, primarily driven by formulary exclusions in more recent years, fits this narrative.

Conclusion

Medicare formularies became increasingly narrow during 2011–20, such that in 2020, beneficiaries' access to drugs in unprotected classes was restricted either through formulary exclusions or prior authorization or step therapy requirements on an average of 40 percent of available compounds. Increasing reliance on formulary exclusions, a particularly blunt tool for restricting patients' access to therapies, is especially striking, as it prevents all but the few who successfully appeal or can afford to pay out of pocket from getting a restricted medication if it is prescribed by their physician. Our analysis could not clearly determine whether these practices are used to ensure that care is delivered cost-effectively or whether these restrictions benefit pharmacy benefit managers' bottom lines while harming patients. The existence of greater restrictions among stand-alone prescription drug plans compared with MA-PD plans certainly suggests that these restrictions could result in higher medical expenditures, which MA-PD plans bear but stand-alone plans do not; this is an important area for further research. ■

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NOTES

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3 An Office of Inspector General report found that Medicare beneficiaries and providers appealed only 1 percent of preauthorization and payment denials, but among the denials that were appealed, Medicare Advantage organizations fully or partially overturned 73 percent of their own denials. See Murrin S. Some Medicare Part D beneficiaries face avoidable extra steps that can delay or prevent access to prescribed drugs [Internet]. Washington (DC): Department of Health and Human Services, Office of Inspector General; 2019 Sep [cited 2024 Feb 5]. (Report No. OEI-09-16-00411). Available from: https://oig.hhs.gov/oei/reports/oei-09-16-00411.pdf

4 Drugs are commonly offered in different forms (for example, tablet, solution, and cream) and might not be perfect substitutes. We took a conservative approach in assuming that if any form of a drug was available in any form, then it was considered "on formulary" in the plan.

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7 In addition to prior authorization or step therapy and formulary exclusions, pharmacy benefit managers and health plans may impose quantity limits on prescription drugs to ensure that prescribed quantities are consistent with clinical dosing guidelines and medical literature. Quantity limits are commonly applied to drugs with abuse potential (for example, opioids and benzodiazepines) and high-cost therapies of uncertain or heterogeneous clinical benefit. Because quantity limits do not restrict patients from filling an initial prescription, we did not evaluate their use in this study.

8 Weighting by sales across all plans would introduce endogeneity, as plans that require prior authorization or step therapy or that exclude a compound from coverage would naturally spend less on it. We therefore computed weights for each compound based on average per member per year spending in plans that did not impose any restrictions on its use—that is, no prior authorization or step therapy or exclusion.

9 To access the appendix, click on the Details tab of the article online.

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